Run report: itemOJ-run-20261018T2132Z

Descriptive record of one recorded run of examples/n2_routes.rs, mode run (Parts O/J) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, 4 threads); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit e8568184d74fa862a5ea28e3a52285b249811958
cmd: n2 research/weil_descent_angles_20261018/results/itemOJ-run-20261018T2132Z run (N2_THREADS=4, sizes 20..40, 10 targets per instance)
rustc 1.97.0 (2d8144b78 2026-07-07)
4
Sun Oct 18 21:32:29 UTC 2026
post-run (d2edb2ab8): summary.txt, routes_summary.csv and ranking.csv re-derived from this run's routes.csv by `n2_routes <dir> summarize` (commit 570037637; adds the K6 lines); routes.csv
unchanged. Tool then rustfmt-formatted (format only, tests pass).
```

tool_log.txt

```
jobs 18 total 166.7s
```

summary.txt

```
O/J bits 20 arm J c 1: IC/rho ratio in [0.507, 1.529] over 12 cells -> unresolved; steps/p 0.311 (rho 0.327)
O/J bits 20 arm J c 2: IC/rho ratio in [0.510, 1.527] over 12 cells -> unresolved; steps/p 0.307 (rho 0.327)
O/J bits 20 arm J c 4: IC/rho ratio in [0.529, 1.556] over 12 cells -> unresolved; steps/p 0.322 (rho 0.327)
O/J bits 20 arm D c 1: IC/rho ratio in [2.796, 3.935] over 6 cells -> rho cheaper in every cell; steps/p 0.371 (rho 0.327)
O/J bits 20 arm D c 2: IC/rho ratio in [2.731, 3.833] over 6 cells -> rho cheaper in every cell; steps/p 0.358 (rho 0.327)
O/J bits 20 arm D c 4: IC/rho ratio in [3.067, 4.330] over 6 cells -> rho cheaper in every cell; steps/p 0.409 (rho 0.327)
O/J bits 24 arm J c 1: IC/rho ratio in [0.594, 1.729] over 12 cells -> unresolved; steps/p 0.307 (rho 0.346)
O/J bits 24 arm J c 2: IC/rho ratio in [0.640, 1.844] over 12 cells -> unresolved; steps/p 0.351 (rho 0.346)
O/J bits 24 arm J c 4: IC/rho ratio in [0.628, 1.783] over 12 cells -> unresolved; steps/p 0.321 (rho 0.346)
O/J bits 24 arm D c 1: IC/rho ratio in [8.414, 14.038] over 6 cells -> rho cheaper in every cell; steps/p 0.450 (rho 0.346)
O/J bits 24 arm D c 2: IC/rho ratio in [7.362, 12.245] over 6 cells -> rho cheaper in every cell; steps/p 0.391 (rho 0.346)
O/J bits 24 arm D c 4: IC/rho ratio in [8.035, 13.371] over 6 cells -> rho cheaper in every cell; steps/p 0.425 (rho 0.346)
O/J bits 28 arm J c 1: IC/rho ratio in [0.632, 1.825] over 12 cells -> unresolved; steps/p 0.294 (rho 0.429)
O/J bits 28 arm J c 2: IC/rho ratio in [0.730, 2.117] over 12 cells -> unresolved; steps/p 0.357 (rho 0.429)
O/J bits 28 arm J c 4: IC/rho ratio in [0.692, 2.001] over 12 cells -> unresolved; steps/p 0.328 (rho 0.429)
O/J bits 28 arm D c 1: IC/rho ratio in [12.289, 23.919] over 6 cells -> rho cheaper in every cell; steps/p 0.358 (rho 0.429)
O/J bits 28 arm D c 2: IC/rho ratio in [13.899, 27.019] over 6 cells -> rho cheaper in every cell; steps/p 0.409 (rho 0.429)
O/J bits 28 arm D c 4: IC/rho ratio in [12.354, 24.013] over 6 cells -> rho cheaper in every cell; steps/p 0.360 (rho 0.429)
O/J bits 32 arm J c 1: IC/rho ratio in [0.813, 2.386] over 12 cells -> unresolved; steps/p 0.298 (rho 0.361)
O/J bits 32 arm J c 2: IC/rho ratio in [0.843, 2.469] over 12 cells -> unresolved; steps/p 0.309 (rho 0.361)
O/J bits 32 arm J c 4: IC/rho ratio in [0.726, 2.119] over 12 cells -> unresolved; steps/p 0.259 (rho 0.361)
O/J bits 32 arm D c 1: IC/rho ratio in [18.276, 40.019] over 6 cells -> rho cheaper in every cell; steps/p 0.332 (rho 0.361)
O/J bits 32 arm D c 2: IC/rho ratio in [22.633, 49.542] over 6 cells -> rho cheaper in every cell; steps/p 0.412 (rho 0.361)
O/J bits 32 arm D c 4: IC/rho ratio in [17.812, 38.992] over 6 cells -> rho cheaper in every cell; steps/p 0.322 (rho 0.361)
O/J bits 36 arm J c 1: IC/rho ratio in [0.734, 2.179] over 12 cells -> unresolved; steps/p 0.342 (rho 0.498)
O/J bits 36 arm J c 2: IC/rho ratio in [0.746, 2.215] over 12 cells -> unresolved; steps/p 0.347 (rho 0.498)
O/J bits 36 arm J c 4: IC/rho ratio in [0.772, 2.299] over 12 cells -> unresolved; steps/p 0.360 (rho 0.498)
O/J bits 36 arm D c 1: IC/rho ratio in [18.403, 45.115] over 6 cells -> rho cheaper in every cell; steps/p 0.374 (rho 0.498)
O/J bits 36 arm D c 2: IC/rho ratio in [20.955, 51.332] over 6 cells -> rho cheaper in every cell; steps/p 0.425 (rho 0.498)
O/J bits 36 arm D c 4: IC/rho ratio in [19.179, 46.997] over 6 cells -> rho cheaper in every cell; steps/p 0.388 (rho 0.498)
O/J bits 40 arm J c 1: IC/rho ratio in [0.942, 2.822] over 12 cells -> unresolved; steps/p 0.337 (rho 0.388)
O/J bits 40 arm J c 2: IC/rho ratio in [0.945, 2.826] over 12 cells -> unresolved; steps/p 0.339 (rho 0.388)
O/J bits 40 arm J c 4: IC/rho ratio in [0.932, 2.787] over 12 cells -> unresolved; steps/p 0.335 (rho 0.388)
O/J bits 40 arm D c 1: IC/rho ratio in [25.118, 67.119] over 6 cells -> rho cheaper in every cell; steps/p 0.368 (rho 0.388)
O/J bits 40 arm D c 2: IC/rho ratio in [24.775, 66.327] over 6 cells -> rho cheaper in every cell; steps/p 0.359 (rho 0.388)
O/J bits 40 arm D c 4: IC/rho ratio in [26.269, 70.104] over 6 cells -> rho cheaper in every cell; steps/p 0.386 (rho 0.388)
O/J scaling arm R c 0: cost ~ p^0.696
O/J scaling arm J c 1: cost ~ p^0.779
O/J scaling arm J c 2: cost ~ p^0.775
O/J scaling arm J c 4: cost ~ p^0.774
O/J scaling arm D c 1: cost ~ p^1.068
O/J scaling arm D c 2: cost ~ p^1.088
O/J scaling arm D c 4: cost ~ p^1.068
```

```
K6 bits 20: rho steps/p mean 0.3270 (s.e. 0.0287, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3566
K6 bits 24: rho steps/p mean 0.3458 (s.e. 0.0384, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3997
K6 bits 28: rho steps/p mean 0.4286 (s.e. 0.0437, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3998
K6 bits 32: rho steps/p mean 0.3610 (s.e. 0.0483, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3565
K6 bits 36: rho steps/p mean 0.4985 (s.e. 0.0542, 30 runs) vs 0.886*sqrt(ell)/p mean 0.4430
K6 bits 40: rho steps/p mean 0.3882 (s.e. 0.0341, 30 runs) vs 0.886*sqrt(ell)/p mean 0.3997
O/J all answers verified: true
```

ranking.csv (first 12 of 36 rows)

bits	arm	c	cells	ic_cheaper_cells	rho_cheaper_cells	min_ratio	max_ratio	verdict
20	J	1	12	6	6	0.5069	1.5289	unresolved
20	J	2	12	6	6	0.5102	1.5267	unresolved
20	J	4	12	6	6	0.5286	1.5557	unresolved
20	D	1	6	0	6	2.7965	3.9349	rho cheaper in every cell
20	D	2	6	0	6	2.7308	3.8330	rho cheaper in every cell
20	D	4	6	0	6	3.0671	4.3299	rho cheaper in every cell
24	J	1	12	6	6	0.5942	1.7286	unresolved
24	J	2	12	6	6	0.6405	1.8440	unresolved
24	J	4	12	6	6	0.6283	1.7834	unresolved
24	D	1	6	0	6	8.4135	14.0384	rho cheaper in every cell
24	D	2	6	0	6	7.3617	12.2449	rho cheaper in every cell
24	D	4	6	0	6	8.0353	13.3714	rho cheaper in every cell

routes.csv (first 12 of 1260 rows)

bits	instance	p	N	h	ell	target	arm	c	factor_base	steps	relations	b_relations	jac_ops	walk_M	walk_I	walk_native	jac_M	jac_I	la_M	la_I	la_native	eadd_M	eadd_I	verified	seconds
20	0	1091	1190984	8	148873	0	R	0	0	291	0	0	0	49105	3270	599	0	0	0	0	0	12	1	1	0.001
20	0	1091	1190984	8	148873	0	J	1	34	281	151	1	1941	89746	2048	942	87753	2025	0	0	1045	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	1	34	607	212	2	0	322779	7494	2928	0	0	0	0	1594	12	1	1	0.003
20	0	1091	1190984	8	148873	0	J	2	67	184	114	4	1755	81811	1860	724	79867	1837	0	0	688	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	2	67	381	166	6	0	214174	5420	2289	0	0	0	0	1135	12	1	1	0.002
20	0	1091	1190984	8	148873	0	J	4	133	200	104	6	1812	84998	1910	875	82228	1887	0	0	560	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	4	133	369	149	13	0	212847	5377	2372	0	0	0	0	815	12	1	1	0.002
20	0	1091	1190984	8	148873	1	R	0	0	317	0	0	0	49931	3318	673	0	0	0	0	0	12	1	1	0.001
20	0	1091	1190984	8	148873	1	J	1	34	356	182	3	1948	87715	1934	1146	85497	1911	0	0	1264	12	1	1	0.001
20	0	1091	1190984	8	148873	1	D	1	34	259	92	1	0	151137	4086	1345	0	0	0	0	552	12	1	1	0.001
20	0	1091	1190984	8	148873	1	J	2	67	244	123	2	1804	81555	1787	875	79370	1764	0	0	752	12	1	1	0.000
20	0	1091	1190984	8	148873	1	D	2	67	251	99	3	0	150811	4114	1519	0	0	0	0	622	12	1	1	0.002

routes_summary.csv (first 12 of 324 rows)

bits	arm	c	ipr	jac_pricing	native_w	runs	verified	mean_p	mean_cost	mean_rho_cost	ratio_to_rho	mean_steps_over_p	rho_steps_over_p
20	J	1	3M	measured	0	30	1	1081	94476.1	62588.3	1.5095	0.3114	0.3270
20	J	1	3M	measured	0.1	30	1	1081	94711.0	62662.7	1.5114	0.3114	0.3270

20	J	1	3M	measured	1	30	1	1081	96825.1	63331.8	1.5289	0.3114	0.3270
20	J	1	3M	eadd	0	30	1	1081	31726.8	62588.3	0.5069	0.3114	0.3270
20	J	1	3M	eadd	0.1	30	1	1081	31961.7	62662.7	0.5101	0.3114	0.3270
20	J	1	3M	eadd	1	30	1	1081	34075.7	63331.8	0.5381	0.3114	0.3270
20	J	1	2log2p_M	measured	0	30	1	1081	128073.1	122390.9	1.0464	0.3114	0.3270
20	J	1	2log2p_M	measured	0.1	30	1	1081	128308.0	122465.2	1.0477	0.3114	0.3270
20	J	1	2log2p_M	measured	1	30	1	1081	130422.1	123134.3	1.0592	0.3114	0.3270
20	J	1	2log2p_M	eadd	0	30	1	1081	65911.6	122390.9	0.5385	0.3114	0.3270
20	J	1	2log2p_M	eadd	0.1	30	1	1081	66146.5	122465.2	0.5401	0.3114	0.3270
20	J	1	2log2p_M	eadd	1	30	1	1081	68260.6	123134.3	0.5544	0.3114	0.3270

Graph

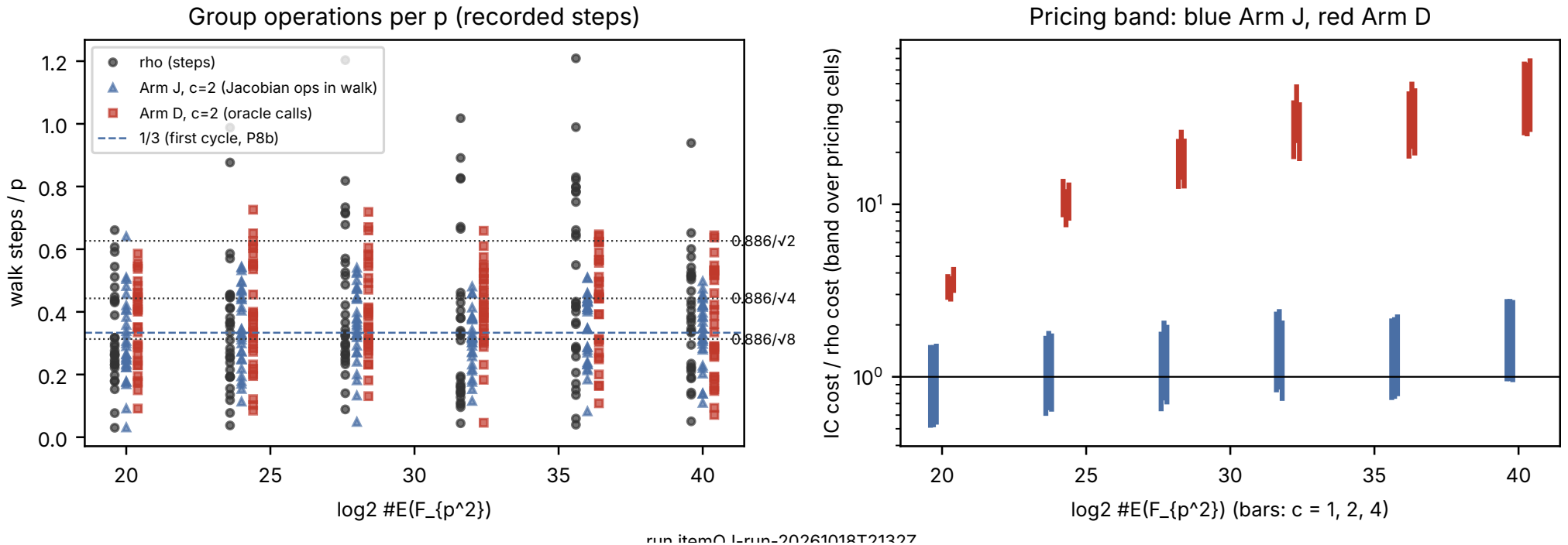


Figure 1: Parts O/J: walk steps per p by arm (left) and the IC/rho cost-ratio band over pricing cells (right); 30 verified runs per arm and size.